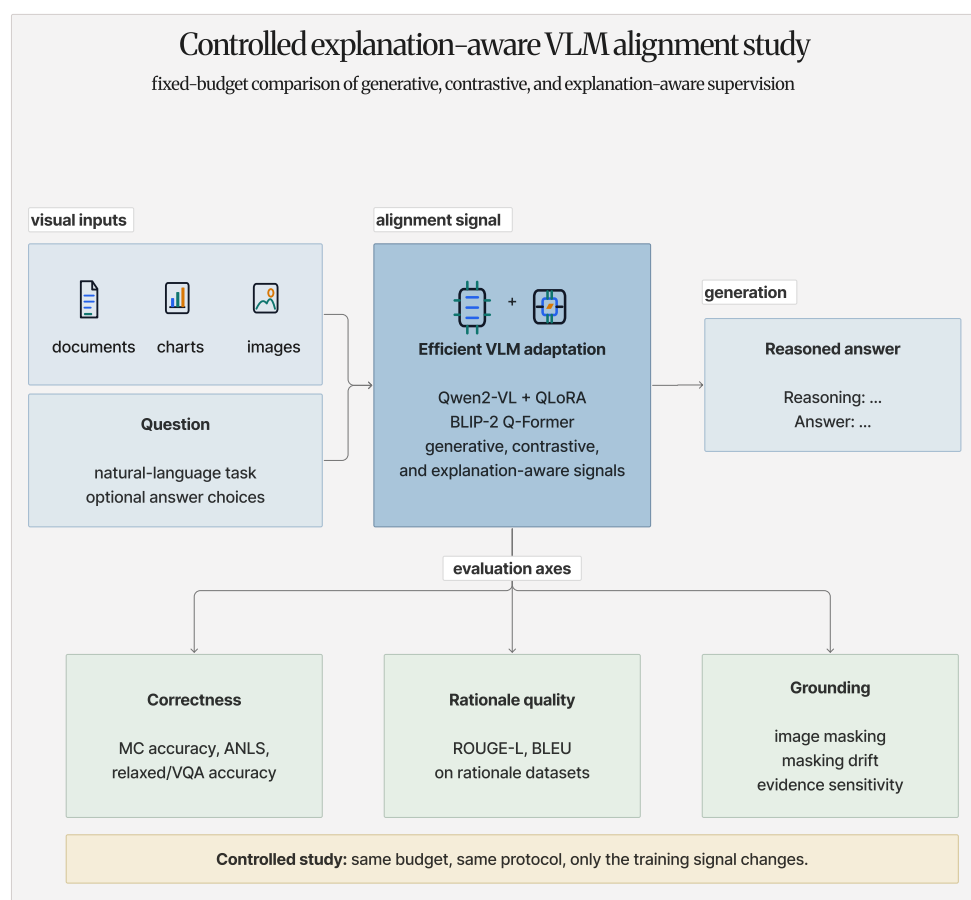


Graphical Abstract

Cross-Modal Generative Alignment for Document, Chart, and Image Reasoning with Explanation-Aware Vision–Language Models

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Highlights

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- The study runs zero-shot, answer-only, rationale-generative, explanation-aware, and BLIP-2 contrastive-enhanced controls under the same single-GPU environment.
- In rationale-bearing datasets, the explanation-aware training does help, but it’s not the best; answer-only controls are strong.
- BLIP-2 contrastive-enhanced fine-tuning yields only a small gain in ScienceQA with BLIP-2 over generative fine-tuning, and the retrieval diagnostic are very close to frozen and random controls.
- Evidence-masking drift shows evidence sensitivity of the model, so the answer gains, rationale quality, and grounding claims are kept separate.

Cross-Modal Generative Alignment for Document, Chart, and Image Reasoning with Explanation-Aware Vision–Language Models

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Abstract

A vision-language model (VLM) is able to read a chart, document, scanned form, or natural image and answer questions about it. But how do we know if the answer is actually derived from the image itself, rather than being a lucky guess on a question based on the model’s remembered language patterns? The correctness of the answer itself is not enough if the reasoning for that answer is not grounded in the visual evidence of the image. Charts and documents are a good example of this problem, because to answer the question correctly, the model has to actually ground its reasoning on specific parts of the image, making it easier to spot when the model is looking at the wrong thing. Although recent work has made progress on this issues, there was no isolation of the training objective as the main variable, with all other factors such as the model, the data, and the budget held constant. Contrastive, generative, and explanation-based methods have all been proposed, but they have not been directly compared under the same conditions, so it is not clear what the objective itself contributed to the final performance. In this paper, we keep the pipeline fixed and change only the training signal, so we can see what the objective does to answer accuracy, explanation quality, and evidence sensitivity when the model, the data, and the compute budget are all held steady. We compare five settings head to head rather than treating them as unrelated systems: BLIP-2 generative fine-tuning, BLIP-2 contrastive-enhanced fine-tuning, Qwen2-VL under answer-only, rationale-generative, and explanation-aware supervision. All the runs use the same resource-constrained single-seed protocol, and the same eval-

uation code as much as possible, across ScienceQA, A-OKVQA, ChartQA, DocVQA, and a VQAv2 subset. The picture that emerges is mixed. On ScienceQA, Qwen2-VL-2B reaches 0.800 accuracy with answer-only supervision, 0.780 in both rationale-generative and length-aware explanation control settings, and 0.765 with fixed-alpha explanation-aware training. On A-OKVQA, the same model reaches 0.830 with answer-only supervision, 0.790 in both rationale-generative and length-aware explanation control training, and 0.795 with fixed-alpha explanation-aware training. ChartQA, DocVQA, and the VQAv2 subset have no gold rationales, so those runs fall back to answer-only controls; they lift the native headline metrics, but they do not test rationale supervision at all. BLIP-2 barely moves: the contrastive objective gives a small gain compared to the generative one on ScienceQA 0.480 vs 0.475, but the retrieval diagnostic stays near frozen and random controls in both cases. Scale is the exception. With a fixed explanation-aware objective on ScienceQA, Qwen2-VL-7B reaches 0.885 while the 2B version reaches 0.765. The masking analysis tells us whether the model relies on the image, not whether its explanation is faithful. Overall, the results do not show a clear win for explanation-aware training. The rationale supervision can help on a single-GPU budget, but the gain is easy to mis-credit to scale or tuning. Hence, it only stands up when answer-only controls, objective balance, model scale, transfer and evidence sensitivity are each reported separately.

Keywords: vision-language models, cross-modal alignment, explanation-aware training, chart and document understanding, faithfulness, parameter-efficient fine-tuning

1. Introduction

Vision-language models (VLMs) are becoming a default tool for answering questions based on the content of images, documents, and charts. The process looks simple: you give it an image and a question, and it will respond with an answer the image actually supports. What is considered as evidence depends entirely on the picture. In a document it might be sitting in a line of a body text, in a table, or in a stamped region of the scanned page. In a chart it might be in an axis label, the relative height of the bars, a colour, or a trend that is not written down explicitly in words. In science and commonsense visual question answering (VQA), it is often not in the image at all, forcing the model to deduce the answer from the background knowledge

it has already learned. Across every one of these, a correct-looking answer is not enough if the reasoning for that answer is not actually based on what is visible in the image. The VLMs are very good at generating the right-looking answers while looking at the wrong thing. That is why in this paper we do not rely on the answer alone, but also check the quality of the explanation and the sensitivity of the answer to visual evidence.

The issue behind all of this is a cross-modal alignment problem: how do we connect the visual signal to the language model in a way that actually makes the two talk to each other? It turns out that the quality of that wiring depends on many factors, such as the visual resolution, the data mixture, the connector design, the instruction tuning, and the adaptation method [1, 2, 3]. Open model families are very helpful, because backbones such as Qwen2-VL, BLIP-2, and LLaVA-OneVision are strong enough out of the box, while still being adaptable on an academic budget [4, 5, 6]. At the same time, document and chart studies show that general visual instruction following is not enough when it comes to text-heavy and structured images [7, 8, 9, 10].

Furthermore, there is also a budget reality to consider. Most students and small labs cannot afford full fine-tuning of a large multimodal model. Methods like DoRA (weight-decomposed low-rank adaptation) make it possible to fine-tune the model by updating only a small portion of the parameters, while keeping the base model frozen or quantised [11]. Hence, in this study, we take the same approach using low-rank adaptation (LoRA) and its quantised variant (QLoRA) instead of full-model fine-tuning [12, 13]. The main goal is not scoring the highest on a leaderboard, but rather to check the model’s performance when only the alignment objective changes while keeping all other factors constant.

1.1. Gap Analysis

The gap is isolation. Most papers change the model, the data, the training size, and the objective all at once. Once the final result is reported, it is hard to tell how much the objective itself contributed. Maybe the objective helped, but it might also be the case that it was simply a bigger model, cleaner data, or more tuning and not necessarily the objective itself.

Moreover, each objective has its own blind spot. Contrastive learning is good at matching an image to a caption, but it does not teach the model to generate an explanation or an answer from the image. Generative learning is good at producing an answer, but it does not necessarily mean that the

model is basing its answer on the correct parts of the image. Explanation-aware training teaches the model to actually write reasoning to come to the answer, but the model could have learned to write a convincing explanation after the fact, which might not actually be the cause of the answer.

Hallucination and faithfulness evaluation has mostly been studied on natural images, where the common test is whether the model names objects that are not there. Charts and documents have recently come into focus too, with benchmarks for chart hallucination, chart grounding, and document-answer groundedness emerging [14, 15, 16]. They are still comparatively underexplored, even though their evidence is more structured and easier to verify. That is exactly what makes them a good test case for grounding.

These gaps are not closed by a new model or benchmark. What is actually missing is a controlled study that holds the model, the data, the budget, and the evaluation procedure constant while changing only the alignment objective, to test not only the answer quality but also the explanation quality and the evidence sensitivity of the model under each objective.

1.2. Research Questions

The main question is to investigate how the resource-constrained vision-language model should be aligned to achieve both accurate and faithful visual reasoning. The following research questions break that down into more measurable parts.

1. RQ1: How have vision-language alignment methods evolved for grounded visual reasoning and explanation generation?
2. RQ2: How does BLIP-2 generative fine-tuning compare with BLIP-2 contrastive-enhanced fine-tuning on visual reasoning?
3. RQ3: Does explanation-aware fine-tuning improve answer accuracy and explanation quality on datasets with gold rationales?
4. RQ4: How does the same fine-tuning framework perform across document, chart, science, commonsense, and natural-image domains?
5. RQ5: How sensitive are the model’s answers to visual evidence under image-masking perturbations?
6. RQ6: How does the selected explanation-aware QLoRA setup scale from Qwen2-VL-2B to Qwen2-VL-7B?
7. RQ7: How well do fine-tuned adapters transfer across visual domains and prompt formats?

83 1.3. Problem Statement

84 The problem is cross-modal alignment in a resource-constrained environ-
85 ment. Given the same pretrained vision–language model, we analyse the
86 effect of different training objectives on answer accuracy, explanation qual-
87 ity, and evidence sensitivity. The model, the data format, the optimiser, and
88 the evaluation process are kept constant wherever possible, leaving only the
89 training signal as the variable. The BLIP-2 pair reuses one Q-Former train-
90 ing setup. The Qwen2-VL generative and explanation-aware runs reuse one
91 QLoRA setup. The goal is not to find the best scoring setup, but to observe
92 whether the chosen objective leads the model in a direction that matters for
93 a more trustworthy reasoning, and to check whether the answer quality, the
94 explanation quality, and the evidence sensitivity move together or separately
95 under each objective.

96 1.4. Novelty of this study

97 The contribution is the controlled comparison, not a new architecture.
98 Instead of introducing a new architecture, we compare existing ones under
99 different training objectives. We consider the faithfulness of the model’s
100 reasoning as a first-class result, rather than concentrating solely on answer
101 accuracy. The paper contributes the following:

- 102 • A comparison of zero-shot, answer-only, BLIP-2 generative, BLIP-2
103 contrastive-enhanced, Qwen2-VL rationale-generative, and Qwen2-VL
104 explanation-aware alignment under the same data and optimisation
105 settings.
- 106 • Evaluation along three separate axes, namely answer quality, explana-
107 tion quality, and evidence sensitivity, across document, chart, science,
108 and commonsense tasks.
- 109 • For multiple-choice reasoning, a generate-then-score protocol is used,
110 so the model is evaluated only after it has produced its own reasoning
111 trace.
- 112 • A direct test of how QLoRA behaves with scale, instead of assuming
113 that full fine-tuning or very large models are the only things worth
114 studying.

115 1.5. *Why this study matters*

116 The usefulness of this study is in the constraints. It runs on a resource-
117 constrained setup that is actually accessible to students and small labs. It
118 also avoids concentrating solely on a single aspect of performance, and in-
119 stead keeps answer accuracy, explanation quality, and evidence sensitivity as
120 separate readings. On documents and charts especially, that separation is
121 particularly important, because the evidence should trace back to a visible
122 region. The whole pipeline is reusable: every dataset is transformed into one
123 shared format, and every experiment is declared in a configuration file, so the
124 code can be used for future comparisons of new objectives, new models, or
125 new datasets, without worrying about hidden differences between runs. And
126 the results are honest rather than overselling: the rationale supervision does
127 help on the science and commonsense runs, but the answer-only controls stay
128 strong, and the BLIP-2 contrastive objective provides only a small answer
129 gain on top of weak retrieval evidence.

130 2. Literature Review

131 The review covers relatively latest works in the areas of vision–language
132 model design, text-rich and document-heavy images, charts and structured
133 visual reasoning, faithfulness and critique, and efficient adaptation. We group
134 it by the parts that actually matter here the most: model design, text-
135 rich images, chart reasoning, faithfulness, datasets, metrics, and low-budget
136 adaptation.

137 2.1. *Recent vision–language model design*

138 Recent vision–language models are not built around a single alignment
139 trick. Their performance is a result of many choices working together, such as
140 the backbone architecture, the visual tokenisation method, the data mixture,
141 the connector design, and the instruction tuning. Since all those factors are
142 usually changed together, it is hard to tell which part actually caused an
143 improvement in the final performance.

144 This is exactly the issues raised by recent ablation studies, which demon-
145 strate that these factors are still poorly separated, and that it is vital to
146 conduct controlled comparisons [1]. MM1.5 also makes the same argument,
147 showing that optical character recognition (OCR) data, synthetic captions,
148 instruction mixtures, and specialised variants can each weigh as much as raw
149 scale in the final performance of a vision–language model [2]. Molmo and

150 PixMo are useful here because they provide open weights and open data that
151 let us actually read the recipe and understand what is going on under the
152 hood [3]. LLaVA-OneVision shows that a single family of models can be used
153 across image and video when the data mixture is chosen with care [6].

154 For this study, the conclusion is simple. If the model, the data, and the
155 budget are all changing together, then the contribution of the objective itself
156 is hard to tell. That is why we need a controlled comparison, where the
157 model, the data, and the budget are held constant while only the training
158 signal changes.

159 Open model families are also important, because they enable us to ac-
160 tually run the controlled comparison on a practical budget. Qwen2-VL is
161 designed to work with dynamic image resolution and text-heavy visual in-
162 puts such as documents, charts, and diagrams [4]. BLIP-2’s Q-Former lets us
163 test an image–answer contrastive loss while keeping the larger vision encoder
164 and language model frozen [5]. We use these open models because they make
165 controlled, parameter-efficient experiments feasible under limited resources.

166 2.2. *Text-rich and document-heavy images*

167 Documents are difficult for a specific reason. The answer in this case is
168 not in the whole image, but rather in a small part of it, such as a line of text, a
169 table cell, or a layout region. Hence, the task is not just an image recognition
170 problem. The model has to read the document and find the evidence in the
171 right place, which is a different skill than naming objects in a natural image.

172 Recent work shows that text-rich image reasoning is moving toward mul-
173 timodal large language models, especially for OCR-heavy and layout-heavy
174 inputs [7]. DocLLM is a good example of this trend. It encodes the document
175 layout directly into the generative document-understanding problem, which
176 is a step toward more faithful reasoning on documents [8]. OmniDocBench
177 focuses on the coverage angle instead, scoring diverse PDF parsing cases
178 with detailed annotations, which is a step toward more structured evalua-
179 tion of document understanding [17]. In this study, we use DocVQA as a
180 test case for document understanding, because it is a well-known benchmark
181 where the answer is usually located on scanned text, layout, and local visual
182 context [18].

183 Our goal is not to build a new document model. We want to see how
184 much the training objective itself contributes to the model’s ability to read
185 documents and answer questions about them.

2.3. Charts and structured visual reasoning

Charts create a different kind of problem. In this case, the model has to read the labels, understand the axes and tick marks, and determine the relationships between the visual elements, such as bars, lines, or points. ChartInstruct and ChartGemma demonstrate that instruction tuning on chart images can help in both chart understanding and reasoning [9, 10]. ChartMimic shows the same point by training a model to generate code for that chart, which makes the model understand the chart in a more structured way [19]. ChartMuseum shows that current large visual-language models still perform far poorer than humans on chart reasoning [20]. MathVista, MMMU-Pro, and MMReason extend this concern to mathematical and multi-step reasoning, which is often required for chart understanding [21, 22, 23].

This is why we use ChartQA, ScienceQA, and A-OKVQA in the same study [24, 25, 26]. Together, they cover structured chart reasoning, science reasoning, and knowledge-heavy visual reasoning.

2.4. Faithfulness, critique, and efficient adaptation

Explanation-aware training has a subtle problem. The model might learn to write an explanation that looks convincing, but that is not actually based on the image and is not the real cause leading to the answer. Recent studies approach this problem from different angles. MMMU tests expert-level multimodal reasoning across many visual formats, to see if VLMs can do expert-style visual reasoning and not just simple image captioning [27]. MMStar checks whether a model actually needs the image to answer the question or whether the answer can be derived from the learned knowledge alone [28]. HallusionBench focuses on cases where language hallucination and visual illusion appear together in LVLM responses [29]. Hallusination-augmented contrastive learning turns a hallucination into a contrastive signal, to teach the model to avoid it [30]. VISCO and Critic-V evaluate the model’s ability to critique, correct, and detect errors in its own output [31, 32]. These works suggest that faithfulness should be measured separately from answer accuracy.

Efficient adaptation is also important, because it allows us to test the effect of the training objective without needing to fine-tune the entire model. DoRA is a good example of this approach, which demonstrates that we can actually adapt the model by updating only a small portion of the parameters [11]. Reason-RFT points to another direction that is relevant here, where post-training specifically rewards visual reasoning [33]. In this paper,

we stay with supervised objective comparisons, but the concern is similar. We evaluate the model as a reasoning system, rather than only as an answer generator.

Table 1 summarises the relevant works and the gaps we address in this study. It is not meant to be a comprehensive review of the field, but rather a snapshot of the recent trends and how they relate to our controlled comparison.

Table 1: Representative related work and how it connects to the gap studied here. A check mark means the line of work directly covers the property, a cross means it is not part of the main scope, and a tilde means partial coverage.

Work	Main focus	Generation	Structured VQA	Rationales	Efficient FT	Connection to this study
Recent VLM construction [1, 3, 6, 2]	Model and data recipe	✓	~	~	~	Shows that model design and training data must be controlled before we can attribute gains to the objective.
Open VLM backbones [4, 5]	Adaptable model families	✓	~	~	~	Provide practical backbones for controlled experiments under a fixed compute budget.
Text-rich document work [7, 8, 17]	Documents and PDF parsing	✓	✓	~	~	Motivates the document setting, but usually does not compare training objectives under one fixed backbone.
Chart reasoning work [9, 10, 20, 19]	Chart understanding	✓	✓	~	~	Shows that chart reasoning is still difficult and needs structured evaluation beyond ordinary image QA.
Recent reasoning benchmarks [21, 27, 28, 22, 23]	Harder multi-modal evaluation	~	✓	~	✗	Motivate evaluation on visual, mathematical, expert-level, and multi-step reasoning tasks.
Faithfulness and critique [29, 30, 31, 32]	Reliability of reasoning	~	✓	✓	✗	Support the need to measure hallucination, critique, and evidence use separately from answer accuracy.
DoRA / Reason-RFT [11, 33]	Efficient and reasoning-focused adaptation	~	✗	~	✓	Motivate low-resource adaptation and explicit reasoning-oriented post-training.
This study	Controlled explanation-aware alignment	✓	✓	✓	✓	Compares objective choices directly and reports answer accuracy, rationale quality, and evidence sensitivity together.

A common pattern emerges from this review. Earlier work on multimodal alignment focused mainly on learning image–text representations, or on generating an answer from the image and the question. More recent work has started to pay attention to the faithfulness of the model’s reasoning, and to the quality of the explanations it generates. However, it is still unclear how much the training objective itself contributes to the model’s performance. In this study, our goal is to fill that gap by conducting a controlled comparison of different training objectives while keeping all other factors constant.

3. Methodology

The main rule in this methodology is to keep the pipeline unchanged and modify only the training objective. Each dataset is transformed into a

241 uniform format. Then, the model is loaded, QLoRA adapters are applied if
242 needed, the model is trained with the chosen objective, and evaluation is per-
243 formed using the same generate-then-score protocol for every run. Figure 1
244 shows the full pipeline.

245 We intentionally keep the pipeline simple and controlled, because the goal
246 is to isolate the effect of the training objective, not to find the best possible
247 score on each dataset. Every experiment is defined through a configura-
248 tion file that declares the dataset, the backbone, the objective, the adapter
249 settings, the metrics, and the seed. The same implementation is used for
250 baseline evaluation, fine-tuning, and final evaluation, reducing the risk of
251 hidden differences affecting the results.

252 3.1. Dataset

253 The experiments use five public datasets. DocVQA represents document
254 understanding, where the answer is usually based on scanned text, layout and
255 local regions of the image [18]. ChartQA is used for chart reasoning, in which
256 the answer might depend on labels, axes, values, or relationships between
257 elements in the chart [24]. ScienceQA provides multimodal science questions
258 with multiple-choice answers and explanations, which is used for rationale
259 supervision [25]. A-OKVQA adds commonsense questions and rationales
260 over natural images, which is also used for rationale-supervised training [26].
261 VQAv2 is used as a general natural-image control, where the answer is often
262 based on object recognition and simple relationships in the image [34].

263 There is one important detail about the VQAv2 subset. The experiment
264 uses the configured `Erland/VQAv2-sample` source, which has 900 training
265 examples and 100 evaluation examples. Hence, we report it as a VQAv2
266 subset result, not as a full VQAv2 benchmark score.

267 All the datasets are transformed into a uniform format, where each ex-
268 ample is comprised of an image, a question, an answer, and optionally a
269 rationale. If the dataset does not have gold rationales, the rationale field is
270 left empty, and the training falls back to answer-only supervision.

271 3.2. Detailed methodology

272 Figure 2 shows the model-level setup. All backbones are loaded in the
273 same way, and the same training code is used for every run, so that we don’t
274 have to design a separate implementation for each model or each objective.
275 In case of Qwen2-VL, the model is loaded in 4-bit precision and trained with
276 rank-16 QLoRA adapters [12, 13]. For the BLIP-2 contrastive run, the vision

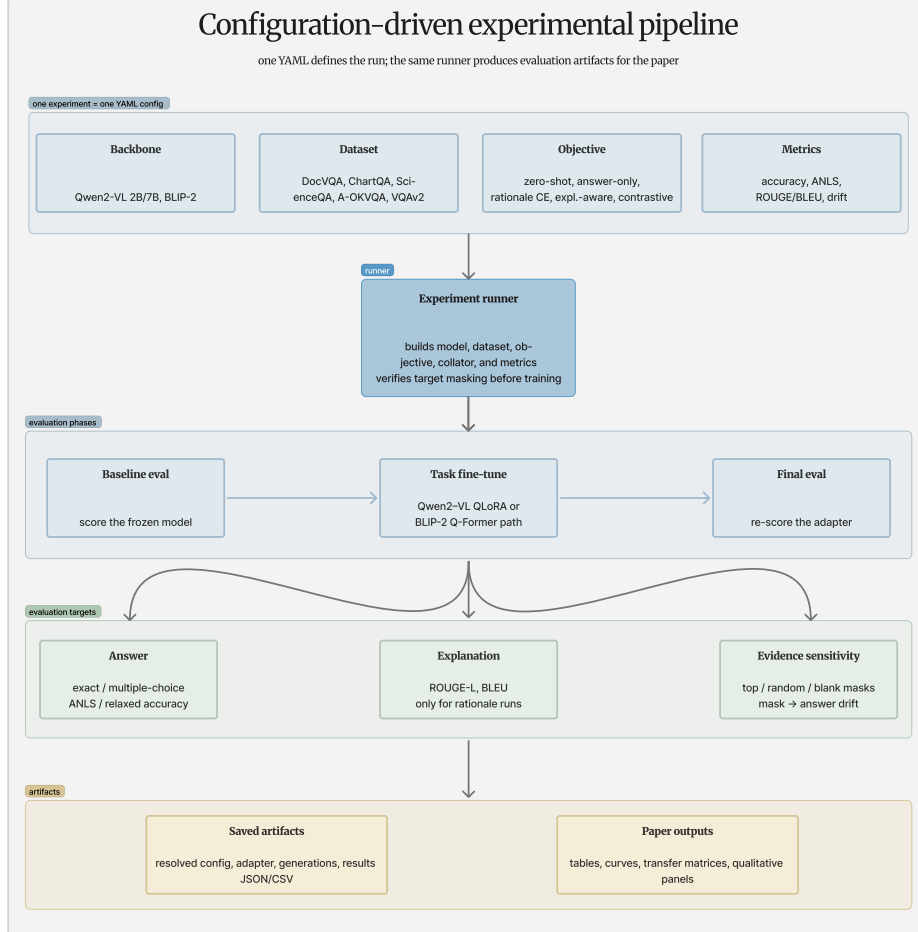


Figure 1: Overall workflow of the proposed explanation-aware alignment study. Datasets are normalised into a shared multimodal example format. Qwen2-VL runs are adapted with QLoRA, BLIP-2 runs train the Q-Former-side alignment path, and each objective is evaluated through answer, rationale, and evidence-sensitivity metrics.

277 encoder and the OPT language model are frozen, while the Q-Former-side
 278 components, the language projection, and a small answer-projection head
 279 are trainable.

280 The difference between the variants is only in the supervision signal. The
 281 baseline is evaluated before the model fine-tuning. The generative vari-

Table 2: Benchmarks used in the study. The sizes are rounded public benchmark sizes before filtering by image availability, answer type, or rationale availability. The completed experiments use capped train and evaluation splits so that every run fits the same practical compute budget.

Dataset	Domain	Approx. size	Answer type	Gold rationale	Primary metric
DocVQA [18]	Document	50K	short text span	no	ANLS
ChartQA [24]	Chart	32K	numeric / short text	no	relaxed accuracy
ScienceQA [25]	Science image	21K	multiple choice	yes	accuracy
A-OKVQA [26]	Commonsense image	25K	multiple choice / short text	yes	accuracy
VQAv2 subset [34]	Natural image	900/100 run split	open ended	no	VQA accuracy

Table 3: Input construction examples. The complete token, label, span, and collator traces are saved as pipeline artifacts; this table gives the paper-level view of what each prompt family means.

Prompt family	Raw fields	Prompt seen by model	Supervised target	Supervised spans
Answer-only	image, question, answer	Question: ... plus choices when available	Answer: <answer>.	answer tokens only
Rationale-generative	image, question, answer, gold rationale	Question: ... plus choices when available	Reasoning: <rationale> Answer: <answer>.	full target under uniform CE
Explanation-aware	image, question, answer, gold rationale	Question: ... plus choices when available	Reasoning: <rationale> Answer: <answer>.	rationale and answer spans weighted separately
Answer-only fall-back	image, question, answer, no gold rationale	Question: ...	Answer: <answer>.	answer tokens only

ant is trained to produce the answer, given the image and question. The contrastive-enhanced variant adds an extra image–answer loss, which pulls the Q-Former pooled image representation closer to the correct answer and pushes it farther from the other answers in the batch. The explanation-aware variant trains the model to generate a rationale first, then the answer, and it weights the rationale and answer spans separately in the loss.

For the generative training, the prompt and targets are concatenated into one sequence, and the loss is computed only over the target tokens. The loss mask is constructed after multimodal encoding, so that the image placeholder tokens included by the processor do not shift the target span off position. This is achieved by measuring the prompt length with the same processor that encodes the full sequence, and masking out that many tokens from the start of the loss.

For contrastive objective, the same span information is used. Only the answer tokens are pooled into the InfoNCE text representation [35]. The rationale tokens still receive the generative loss, but they are not used in the contrastive loss. Equation 1 weights these two parts using α . When $\alpha = 1$, only the answer span is weighted, even if the target may still contain the rationale before the answer. That is why the true answer-only control is reported separately with a target template that contains only **Answer:**. When

302 $\alpha = 0$, only the explanation span is supervised. The intermediate values
 303 are used to test the interaction between answer and rationale supervision,
 304 whether they help each other or compete.

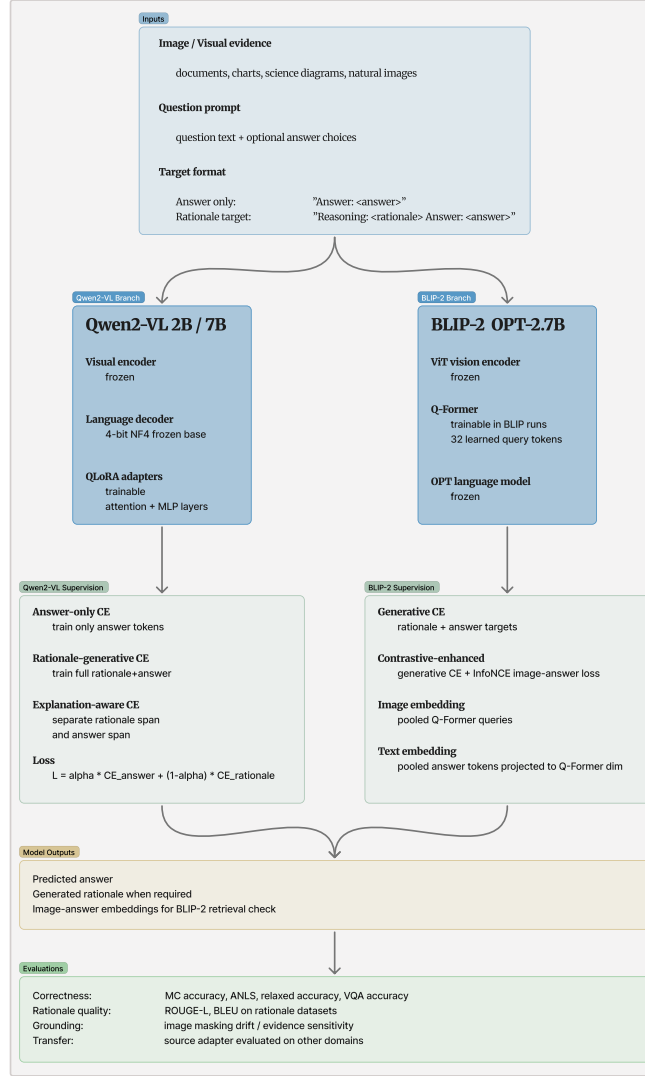


Figure 2: Model-level architecture for explanation-aware cross-modal alignment. Visual evidence and the question prompt are encoded into a multimodal token stream. The Qwen2-VL path uses trainable QLoRA adapters and separate rationale and answer supervision spans. The BLIP-2 path keeps the vision tower and OPT language model frozen and trains the Q-Former-side generative or contrastive alignment components.

$$\mathcal{L} = \alpha \mathcal{L}_{\text{answer}} + (1 - \alpha) \mathcal{L}_{\text{explanation}} \quad (1)$$

Here, $\mathcal{L}_{\text{answer}}$ is the masked cross-entropy loss on the answer tokens, and $\mathcal{L}_{\text{explanation}}$ is the same loss on the explanation tokens. The coefficient α controls the relative weight of the answer and explanation supervision.

We use two ways of setting α . In fixed-alpha mode, α is chosen explicitly in the experiment configuration serving as an ablation control. In length-aware mode, the effective answer weight is computed from the number of supervised answer and explanation tokens:

$$\alpha_{\text{eff}} = \frac{\lambda n_{\text{answer}}}{\lambda n_{\text{answer}} + n_{\text{explanation}}}, \quad (2)$$

where n_{answer} and $n_{\text{explanation}}$ are the answer and explanation token counts after collation, and λ is the answer-weight multiplier. When $\lambda = 1$, the loss becomes the usual token-level cross-entropy over the full rationale-plus-answer target. Because of this, the fixed-alpha runs are treated as the ablation, while the length-aware run is used as the natural token-weighted control.

For multiple-choice datasets, inference is done in two steps. First, a short reasoning sequence is generated given the image and the question. Then, given that reasoning trace, the model scores each answer option by computing the length-normalised log-likelihood of that option as a continuation of the reasoning. Afterwards, the option with the highest score is selected as the final answer. This avoids relying on fragile string parsing or on whether the model copied an option label correctly. In case of open-ended datasets, the model just generates an answer string, which is evaluated using the dataset’s native metric. When a rationale is generated, it is saved for later use in the rationale quality and evidence sensitivity evaluation.

3.3. Evaluation metrics

Evaluation is divided into three parts: answer quality, rationale quality, and evidence sensitivity. Answer quality is measured using the native metric of the benchmark, which is the primary metric reported on the leaderboard for that dataset. We use the term “headline metric” to refer to this native primary metric, not to some universal metric to be compared directly across datasets. For ScienceQA and A-OKVQA, the headline metric is multiple-choice accuracy. For DocVQA, it is average normalised Levenshtein similarity (ANLS), as per the benchmark convention [18]. For ChartQA, it is

336 relaxed accuracy for numerical answers [24]. For the VQAv2 subset, it is
 337 VQA consensus accuracy, which is computed wherever annotator answers
 338 exist [36, 34].

339 For rationale quality, we use BLEU and ROUGE-L whenever gold ra-
 340 tionales are available [37, 38]. These metrics do not measure faithfulness or
 341 correctness of the rationale, but they do provide a basic check on whether the
 342 generated explanation is lexically similar to the gold rationale, and whether
 343 it captures the same sequence of reasoning steps.

344 Evidence sensitivity is measured in an iterative way. The model’s an-
 345 swer is scored on the original image, and then scored again several times by
 346 masking out different regions of the image. This is a general idea of occlu-
 347 sion sensitivity analysis [39], adapted to the question-answering setting. The
 348 drift in the answer score when a region is masked out indicates how much
 349 the model’s answer relies on that region as evidence. That evidence-masking
 350 drift is shown in Equation 3. For qualitative inspection, we compare the
 351 regions with high evidence-masking drift to the content named in the gener-
 352 ated rationale, to see if the model is actually naming the evidence it relies
 353 on.

$$\text{drift}(r) = s(a \mid I) - s(a \mid I \setminus r) \quad (3)$$

354 where $s(a \mid I)$ is the length-normalised log-likelihood of the gold answer
 355 a given image I , and $I \setminus r$ is the same image with region r occluded. A
 356 large positive drift says the masked region mattered. A drift near zero says
 357 the opposite: the answer probably leaned on language priors, a shortcut, or
 358 non-visual context. The absolute scale does not travel across datasets, since
 359 answer length, prompt family, image resolution, and task format all bend the
 360 likelihood scores. The split it captures is the familiar one between a useful
 361 explanation and a faithful one. It also speaks to recent hallucination and
 362 critique work, where a model happily describes or relies on visual content
 363 that simply is not there [29, 30, 31, 32]. That is why evidence sensitivity
 364 sits next to answer accuracy here, rather than being parked as a post-hoc
 365 example.

366 3.4. Experimental settings

367 Fixed-budget protocol is applied for all the experiments, where every run
 368 is capped to the same number of training examples, the same number of
 369 evaluation examples, and the same number of seeds. For Qwen2-VL, the

370 default setup is AdamW optimizer with a learning rate of 2×10^{-4} , cosine
 371 decay, three percent warm-up, weight decay of 0.01, bf16 compute, and an
 372 effective batch size of 32 through gradient accumulation. The base model
 373 is loaded with 4-bit NormalFloat (NF4) quantisation. QLoRA adapters use
 374 rank 16 and scaling factor 32 [12, 13]. Qwen2-VL-2B-Instruct is the main
 375 backbone because it is strong enough while still being small enough to fit the
 376 available budget [4]. The BLIP-2 experiments use Salesforce BLIP-2 OPT-
 377 2.7B, where only Q-Former-side components are trainable, while the vision
 378 tower and the OPT language model are frozen [5]. Qwen2-VL-7B with the
 379 same QLoRA setting is used for the scale test.

380 The split, filtered example count, seed, objective, adapter configuration,
 381 and evaluation metrics are recorded for each run. A split-and-leakage au-
 382 diting is performed for every run, to check the overlap between the train and
 383 evaluation identifiers. The overlap between train and evaluation in the final
 384 artifact is zero. The only non-standard split is DocVQA. The loaded source
 385 provides only validation and test. That is why the experiment run uses
 386 `validation[:80%]` for training and `validation[80%:]` for evaluation, with
 387 native-ID overlap checking. That audit shows zero native-ID overlap, but it
 388 does not guarantee that the train and evaluation sets are completely disjoint
 389 in terms of visual content, because the dataset contains multiple questions
 390 per document. For that reason, the DocVQA score is viewed as an internal
 391 adaptation result, not as a benchmark claim.

392 Approximate uncertainty estimates are reported for the accuracy-style
 393 headline metrics. Since most evaluation sets are 200 examples and only a
 394 small number of seeds fit the budget, very small gaps like 0.790 versus 0.795
 395 are treated as observed differences rather than statistically significant ones.
 396 Table 4 gives the default configuration used for the Qwen2-VL runs.

397 3.5. *Experiment ledger*

398 Before reporting the results, a summary of what each group of runs is
 399 meant to test is shown in Table 5. The key boundary is availability of
 400 gold rationales. ScienceQA and A-OKVQA have gold rationales, meaning
 401 they support rationale-generative and explanation-aware supervision. On the
 402 other hand, ChartQA, DocVQA, and VQAv2 do not provide the rationale
 403 target, so they are considered as answer-only fallback runs for the domain
 404 and transfer analyses.

Table 4: Default Qwen2-VL QLoRA fine-tuning configuration. BLIP-2 contrastive runs use the separately reported frozen-vision/frozen-LLM Q-Former training setup. Any run that changes these values is reported separately in the experiment ledger and in the Results section.

Training Configuration	
Adapter	QLoRA (4-bit NF4, rank 16, scaling 32)
Optimizer	AdamW ($\beta_1=0.9$, $\beta_2=0.999$)
Learning rate	2×10^{-4} (cosine, 3% warm-up)
Weight decay	0.01
Effective batch size	32 with gradient accumulation
Precision	bf16 compute
Primary seed policy	one seed per complete configuration

Table 5: Main experiment ledger. The full generated ledger includes exact caps, seeds, prompt variants, alpha mode, and configuration hashes.

Comparison	Runs included	Use in the paper
BLIP-2 objective pair	BLIP-2 OPT-2.7B on ScienceQA: generative and generative+contrastive.	Tests whether the auxiliary contrastive objective contributes anything beyond BLIP-2 generative fine-tuning.
ScienceQA controls	Qwen2-VL-2B on ScienceQA: answer-only, rationale+answer CE, fixed- α explanation-aware sweep, and length-aware control.	Separates answer supervision, rationale supervision, and alpha weighting on a gold-rationale dataset.
A-OKVQA controls	Qwen2-VL-2B on A-OKVQA: answer-only, rationale+answer CE, fixed- α explanation-aware, and length-aware control.	Repeats the rationale-supervision comparison on a second gold-rationale domain.
Fallback domains	Qwen2-VL-2B on ChartQA, DocVQA, and VQAv2 with answer-only fallback prompts.	Reports in-domain adaptation and transfer for datasets without gold rationales; these rows are not used as explanation-aware evidence.
Scale	Qwen2-VL-2B and Qwen2-VL-7B on ScienceQA with the same fixed explanation-aware QLoRA setting.	Checks whether the selected setup changes when the backbone scale increases.
Transfer	Qwen2-VL-2B source adapters evaluated on target-domain prompts across all datasets.	Tests cross-domain behaviour separately from the official in-domain result table.
Masking	Selected Qwen2-VL runs with image-region perturbation and saved visual examples.	Measures evidence sensitivity beside answer accuracy and rationale quality.

405 4. Results

406 The results are reported following the same order as shown in the ex-
407 periment ledger. First, the BLIP-2 pair, which tests whether the auxil-
408 iary contrastive term adds anything on top of generative fine-tuning. Then,
409 ScienceQA and A-OKVQA controls, which separate answer-only, rationale-
410 generative, and explanation-aware supervision. And finally, the last three
411 controls report masking-based visual sensitivity, the Qwen2-VL scale check,
412 and cross-domain transfer. It is important to note that the results should
413 be viewed in the context of the small-budget protocol, not the leaderboard
414 submissions.

415 4.1. Objective-level comparison

416 The objective-level results are shown in Table 6. For BLIP-2 on Sci-
417 enceQA, both objectives yield a slight improvement over the frozen baseline
418 of 0.445. The contrastive-enhanced run reaches 0.480, while the generative
419 control reaches 0.475. But the difference is very small and falls within the
420 range of the approximate uncertainty intervals in Table 7. The same is true
421 for the retrieval diagnostic, where the contrastive-enhanced model has R@1
422 of 0.010, R@5 of 0.020, R@10 of 0.030, and MRR of 0.0307, which is not a
423 meaningful improvement over the generative control.

424 Qwen2-VL-2B shows stronger results overall. Rationale-generative fine-
425 tuning scores 0.780 on ScienceQA and 0.790 on A-OKVQA. Explanation-
426 aware run on A-OKVQA scores 0.795 accuracy, 0.337 ROUGE-L, and 0.087
427 BLEU. That is slightly above the generative control, but the difference is
428 small enough that it should be treated as an observed single-run result, not
429 a proven advantage. The answer-only controls are also strong: 0.800 on
430 ScienceQA and 0.830 on A-OKVQA. The zero-shot column is reported as a
431 common reference for each model and dataset, not as a separate zero-shot
432 run for every prompt style. On ScienceQA, the fixed $\alpha = 0.50$ explanation-
433 aware run scores 0.765, while the length-aware control scores 0.780, matching
434 the rationale-generative control but still below the answer-only run.

Table 6: Objective-level results under the controlled protocol. The zero-shot reference is the shared score for the same model and dataset group before the fine-tuning, not a separately recomputed frozen score for every prompt family. Accuracy is multiple-choice accuracy for ScienceQA and A-OKVQA.

Backbone	Dataset	Objective	Zero-shot ref.	Final	ROUGE-L	BLEU
BLIP-2 OPT-2.7B	ScienceQA	Generative	0.445	0.475	–	–
BLIP-2 OPT-2.7B	ScienceQA	Contrastive-enhanced	0.445	0.480	–	–
Qwen2-VL-2B	ScienceQA	Answer-only	0.400	0.800	–	–
Qwen2-VL-2B	ScienceQA	Rationale+answer CE	0.400	0.780	0.693	0.559
Qwen2-VL-2B	ScienceQA	Explanation-aware, $\alpha = 0.50$	0.400	0.765	0.643	0.493
Qwen2-VL-2B	ScienceQA	Explanation-aware, length-aware	0.400	0.780	0.696	0.563
Qwen2-VL-2B	A-OKVQA	Answer-only	0.355	0.830	–	–
Qwen2-VL-2B	A-OKVQA	Rationale+answer CE	0.355	0.790	0.329	0.083
Qwen2-VL-2B	A-OKVQA	Explanation-aware, $\alpha = 0.50$	0.355	0.795	0.337	0.087
Qwen2-VL-2B	A-OKVQA	Explanation-aware, length-aware	0.355	0.790	0.330	0.083

Table 7: Approximate uncertainty for selected accuracy-style headline metrics. These intervals use the generated normal-approximation standard errors for capped evaluation sets, so they are a caution about small observed differences rather than a replacement for multi-seed testing.

Run	Metric	Eval cap	Score	Approx. 95% CI
BLIP-2 generative ScienceQA	MC accuracy	200	0.475	0.406–0.544
BLIP-2 contrastive ScienceQA	MC accuracy	200	0.480	0.411–0.549
ScienceQA answer-only	MC accuracy	200	0.800	0.745–0.855
ScienceQA rationale+answer CE	MC accuracy	200	0.780	0.723–0.837
ScienceQA explanation-aware, $\alpha = 0.50$	MC accuracy	200	0.765	0.706–0.824
ScienceQA explanation-aware, length-aware	MC accuracy	200	0.780	0.723–0.837
A-OKVQA answer-only	MC accuracy	200	0.830	0.778–0.882
A-OKVQA rationale+answer CE	MC accuracy	200	0.790	0.734–0.847
A-OKVQA explanation-aware, $\alpha = 0.50$	MC accuracy	200	0.795	0.739–0.851
Qwen2-VL-7B ScienceQA, $\alpha = 0.50$	MC accuracy	200	0.885	0.841–0.929

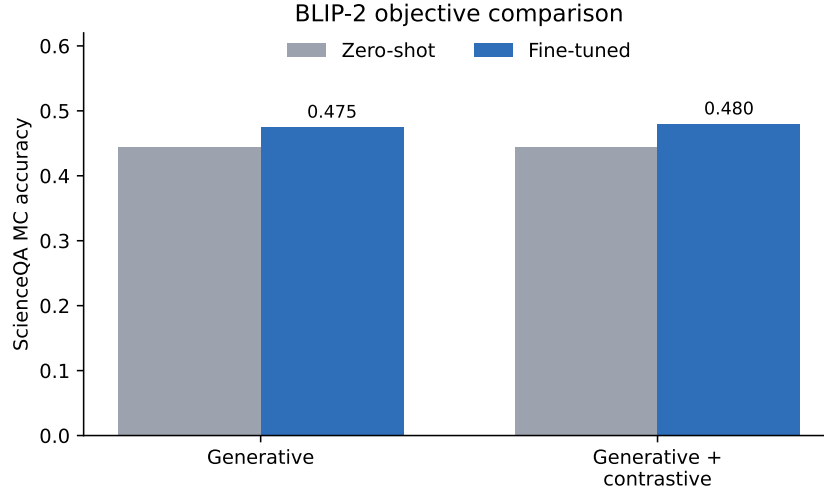


Figure 3: BLIP-2 generative and contrastive-enhanced ScienceQA results. Both runs improve over the frozen baseline, with only a small observed gain from the auxiliary contrastive term.

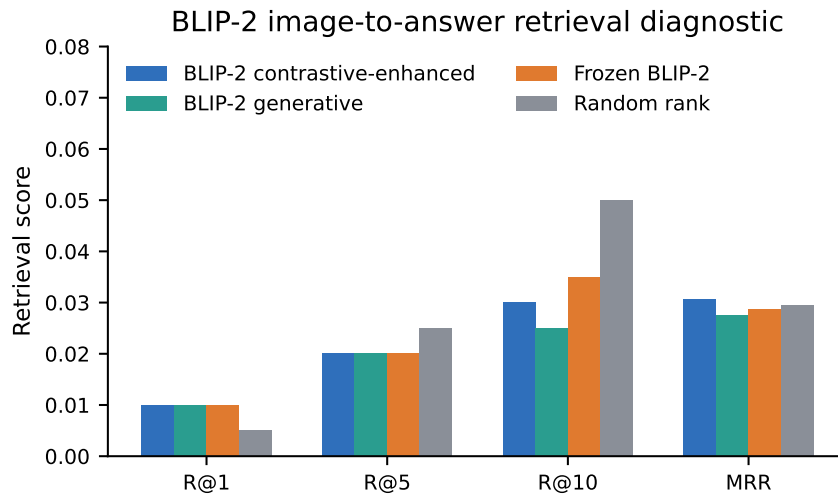


Figure 4: BLIP-2 image-to-answer retrieval diagnostic. The analysis reports frozen BLIP-2, BLIP-2 generative, BLIP-2 contrastive-enhanced, and random-rank controls so the contrastive branch is not interpreted without a reference point.

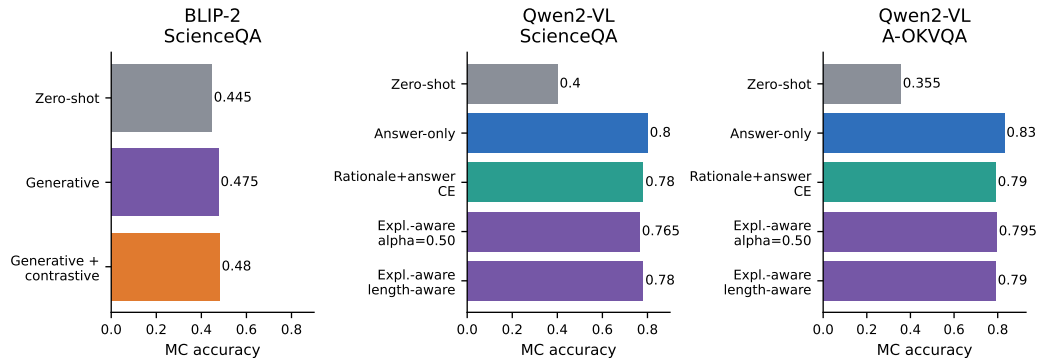


Figure 5: Controlled strategy comparison for BLIP-2 and Qwen2-VL. The figure shows zero-shot, answer-only, rationale-generative, explanation-aware, and contrastive variants where those controls are available.

435 4.2. Effect of explanation-aware training

436 Table 8 and Figure 6 show the ScienceQA fixed- α sweep. Here α is the
 437 answer-span weight from Equation 1. Fine-tuning raises accuracy from the
 438 0.400 zero-shot reference into the 0.690–0.780 range. The best fixed setting
 439 is low answer weight: $\alpha = 0.10$, which reaches 0.780. As α moves toward
 440 the answer-heavy end, accuracy drops to 0.690 at $\alpha = 0.75$ and 0.695 at
 441 $\alpha = 1.00$. The same pattern is observed on the rationale metrics. ROUGE-L
 442 and BLEU also reach the maximum at $\alpha = 0.10$ (0.695 and 0.566), then
 443 decline as α increases. At $\alpha = 1.00$, only the answer span is supervised
 444 inside a target but the rationale part is still included in the target template,
 445 although it is not weighted in the loss. Hence, the rationale metrics are not
 446 computed in that case.

447 The $\alpha = 0.10$ run matches the rationale-generative and length-aware
 448 controls on accuracy. The $\alpha = 1.00$ run should not be confused with a true
 449 answer-only run, because the target template in that case still contains the
 450 rationale part, even though that rationale span is not included in the loss.
 451 The answer-only control on ScienceQA is trained with a target that contains
 452 only the answer part of the template, reaches 0.800, which is above the best
 453 fixed- α run and the rationale-generative control. All of these comparisons
 454 use one seed, 2,000 training examples, and 200 evaluation examples, so they
 455 show observed behaviour rather than final statistical proof.

Table 8: Explanation-aware fixed- α sweep on ScienceQA (Qwen2-VL-2B, QLoRA, 2,000 train / 200 eval, single seed). The coefficient α weights the answer span in Equation 1: $\alpha = 1$ is answer-span-only within the rationale target and $\alpha = 0$ is explanation-span-only. The zero-shot and rationale-generative rows are references. Random-choice accuracy is 0.354. The best fixed α is shown in bold.

Variant (α)	MC accuracy	ROUGE-L	BLEU
Zero-shot (frozen)	0.400	–	–
Rationale+answer CE	0.780	0.693	0.559
$\alpha = 0.00$ (explanation-only)	0.725	0.417	0.299
$\alpha = 0.10$	0.780	0.695	0.566
$\alpha = 0.25$	0.775	0.686	0.548
$\alpha = 0.50$	0.765	0.643	0.493
$\alpha = 0.75$	0.690	0.604	0.443
$\alpha = 1.00$ (answer-span only)	0.695	–	–

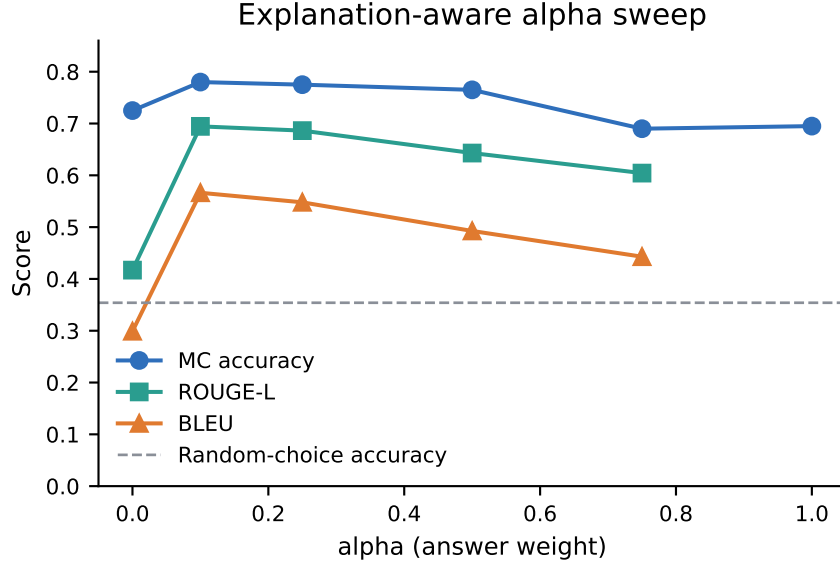


Figure 6: Fixed-alpha explanation-aware sweep on ScienceQA. Answer accuracy peaks at the low-answer-weight setting and declines toward both extremes inside the rationale-plus-answer target. Explanation quality (ROUGE-L, BLEU) peaks at the same point and falls as α increases. Series are distinguished by marker shape and line style; the dashed grey line marks random-choice accuracy. Single seed.

4.3. Domain-level behaviour

Table 9 and Figure 7 represent the results for the selected in-domain runs. Fixed $\alpha = 0.50$ is used for the ScienceQA row, so that it matches the selected domain comparison. The stronger $\alpha = 0.10$ and length-aware settings are reported separately in the sweep above. A-OKVQA rises from 0.355 to 0.795 accuracy against the zero-shot reference. The VQAv2 subset rises from 0.350 to 0.730 VQA accuracy. ChartQA reaches 0.655 relaxed accuracy, with exact match at 0.410. DocVQA reaches 0.905 ANLS and 0.855 exact match. Although the DocVQA score is strong in this run, it should be read carefully, because the examples are mostly short extractive answers, the training setup is answer-only fallback rather than gold-rationale supervision, and the split audit cannot prove document-level disjointness beyond the native IDs exposed by the loader.

Table 9: Domain-level results for the completed Qwen2-VL-2B runs. Each domain uses its native headline metric: multiple-choice accuracy for ScienceQA and A-OKVQA, relaxed accuracy for ChartQA, ANLS for DocVQA, and VQA accuracy for the VQAv2 subset.

Domain	Dataset	Objective	Zero-shot ref.	Final	Secondary metric
Science	ScienceQA	Explanation-aware, $\alpha = 0.50$	0.400	0.765	ROUGE-L 0.643, BLEU 0.493
Commonsense	A-OKVQA	Explanation-aware, $\alpha = 0.50$	0.355	0.795	ROUGE-L 0.337, BLEU 0.087
Charts	ChartQA	Answer-only fallback	0.335	0.655	Exact match 0.410
Documents	DocVQA	Answer-only fallback	0.349	0.905	Exact match 0.855
Natural images	VQAv2 subset	Answer-only fallback	0.350	0.730	Exact match 0.350 $\rightarrow$ 0.730

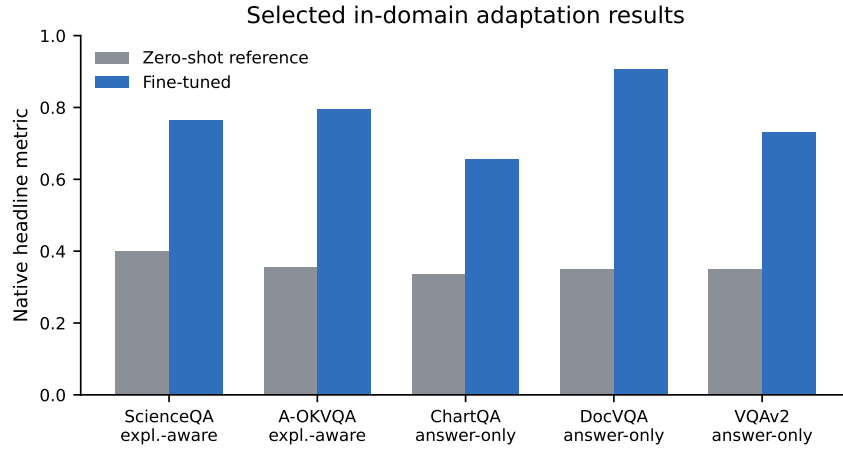


Figure 7: In-domain adaptation results for the selected completed Qwen2-VL-2B domain runs. The comparison mixes rationale-supervised ScienceQA/A-OKVQA rows with answer-only fallback rows for domains without gold rationales, so it should not be read as an explanation-aware effect across all domains.

4.4. Faithfulness and hallucination analysis

Evidence-masking drift is shown in Table 10 and Figure 8. Each value is averaged over 30 evaluation examples. Higher positive drift means that the model produced lower probability for the real answer when the masked region was occluded, meaning that the model relied on that region as evidence for the answer. The strongest evidence sensitivity is observed in the structured tasks: DocVQA at 0.375, ChartQA at 0.224, and VQAv2 at 0.111. ScienceQA and A-OKVQA are much lower, with the ScienceQA answer-only adapter as the main exception at 0.0898.

Low ScienceQA drift is understandable, because many ScienceQA items could be partly answered based on the question and the answer choices using the learned facts, the relevant visual evidence can be spread across the image, and a coarse mask may miss the regions the model actually uses. Explanation

482 supervision does not create a clean drift pattern in this run. The $\alpha = 0.10$
483 setting gives 0.0223, $\alpha = 0.50$ gives 0.0197, and answer-span-only $\alpha = 1.00$
484 gives 0.0149, even though that run has no supervised rationale metric. On
485 A-OKVQA, the explanation-aware run has slightly higher drift than the generative control (0.0271 versus 0.0245), but both intervals are wide. Hence,
486 the evidence-masking results do not demonstrate that the explanation-aware
487 training yields a stronger visual sensitivity than the rationale-generative control.
489

Table 10: Evidence-masking drift by completed run. Higher values indicate stronger sensitivity to masked visual evidence. The intervals are approximate 95% intervals from the generated masking artifact.

Run	Mean drift	Approx. 95% CI	Examples
ScienceQA answer-only	0.0898	0.0197–0.1830	30
ScienceQA rationale+answer CE	0.0217	-0.0100–0.0582	30
ScienceQA $\alpha = 0.10$	0.0223	-0.0080–0.0568	30
ScienceQA $\alpha = 0.50$	0.0197	-0.0119–0.0552	30
ScienceQA answer-span-only $\alpha = 1.00$	0.0149	-0.0274–0.0638	30
A-OKVQA rationale+answer CE	0.0245	-0.0236–0.0726	30
A-OKVQA explanation-aware	0.0271	-0.0075–0.0627	30
ChartQA answer-only fallback	0.2237	0.1419–0.3140	30
DocVQA answer-only fallback	0.3749	0.2770–0.4960	30
VQAv2 answer-only fallback	0.1109	0.0274–0.2408	30
Qwen2-VL-7B ScienceQA $\alpha = 0.50$	0.0114	-0.0059–0.0260	30

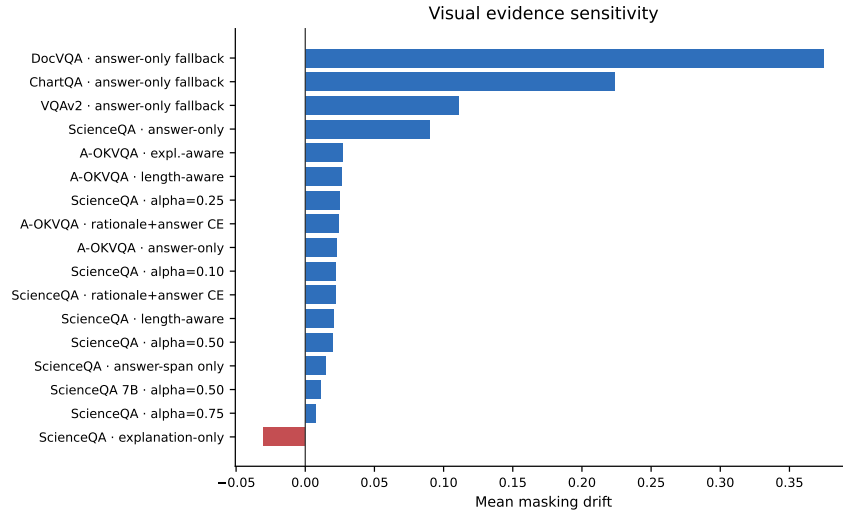


Figure 8: Mean evidence-masking drift across completed runs. Structured document and chart runs show much stronger visual sensitivity than the rationale-rich ScienceQA and A-OKVQA runs, but stronger drift does not automatically imply better answer accuracy.

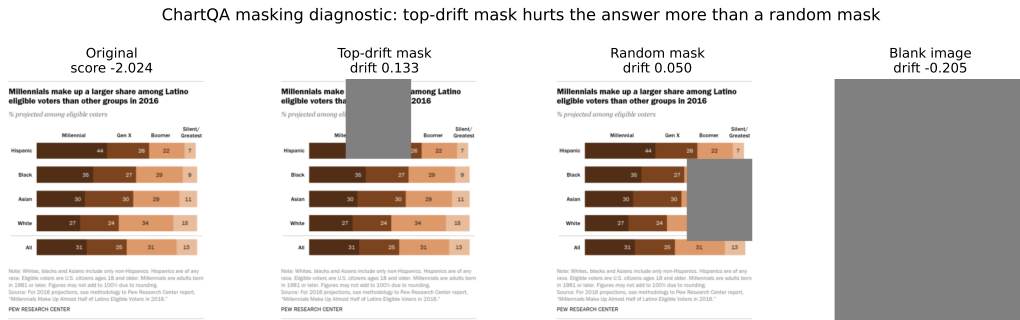


Figure 9: Example masking diagnostic for a ChartQA answer-only fallback run. The pipeline saves the original image, the top-drift mask, a random-region mask, and a blank-image control, together with the full-image and masked answer scores. This figure is used as an evidence-sensitivity example, not as proof that a generated rationale is faithful.

4.5. Efficiency and scale trade-offs

The comparison between Qwen2-VL-2B and Qwen2-VL-7B using the same ScienceQA explanation-aware setting ($\alpha = 0.50$) is shown in Table 11 and Figure 10. The larger model shows a clear improvement in the completed experiments. Accuracy rises from 0.765 to 0.885. ROUGE-L moves

495 from 0.643 to 0.738, and BLEU moves from 0.493 to 0.637. The comparison
 496 is still within the same parameter-efficient setting, because both runs are
 497 fine-tuned using the same QLoRA setup. However, the logs do not include
 498 the exact GPU hours or memory usage for each run, so we treat this as a
 499 scale comparison rather than a full efficiency benchmark.

Table 11: Qwen2-VL scale comparison on ScienceQA with the same explanation-aware objective ($\alpha = 0.50$).

Model	MC accuracy	ROUGE-L	BLEU
Qwen2-VL-2B	0.765	0.643	0.493
Qwen2-VL-7B	0.885	0.738	0.637

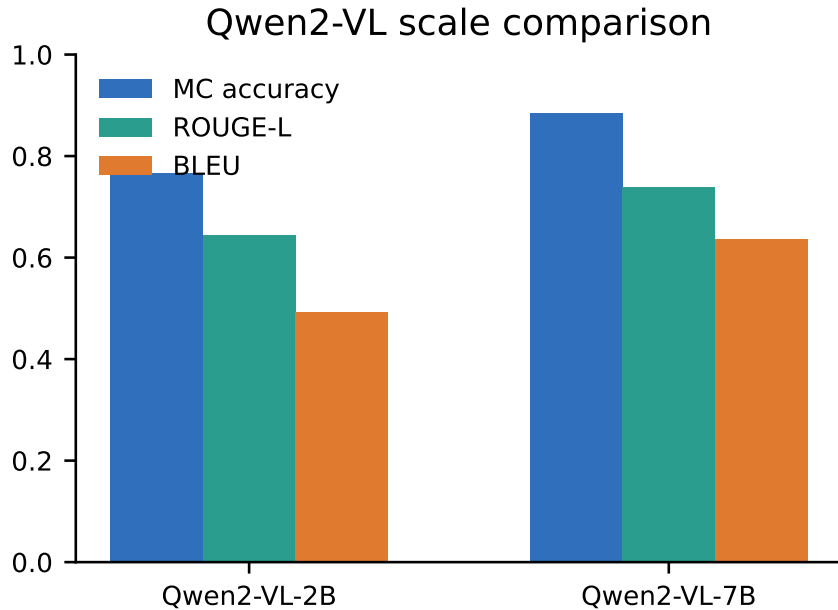


Figure 10: Scale comparison for the fixed ScienceQA explanation-aware setting. Moving from Qwen2-VL-2B to Qwen2-VL-7B improves both answer accuracy and rationale-overlap metrics.

500 4.6. Cross-domain transfer

501 Table 12 and Figure 11 present the transfer matrix from training domain
 502 to evaluation domain. Each column uses the metric of the target dataset,

so values should be compared within a column rather than across a row. The matrix keeps answer-only, rationale-generative, and explanation-aware sources separate. This table is a transfer probe, not a replacement for the in-domain results. Since the transfer evaluator might use a different prompt family than the in-domain evaluation (e.g., answer-only transfer evaluated with an answer-only prompt), the diagonal values may not be exactly the same as the in-domain table.

The transfer results are not explained by the dataset domain alone, meaning that the training objective also influences how well a model transfers. The ScienceQA answer-only adapter transfers well, reaching 0.805 on A-OKVQA, 0.645 on ChartQA, 0.703 on DocVQA, and 0.630 on VQAv2. The ScienceQA rationale-generative and explanation-aware adapters are above random baselines, but they transfer less well to ChartQA, DocVQA, and VQAv2 compared to the answer-only adapter. For A-OKVQA sources, rationale-generative training gives the strongest DocVQA transfer in that group (0.764), while answer-only training gives the strongest VQAv2 transfer (0.700). DocVQA is strongest in-domain at 0.910, and the document-trained adapter also transfers well to ChartQA at 0.685. Overall, transfer depends on both the target domain and the source objective.

Table 12: Cross-domain transfer matrix. Rows indicate the training domain and columns indicate the evaluation domain. Each column uses the target domain’s headline metric, so values are best compared within a column.

Trained on	ScienceQA	A-OKVQA	ChartQA	DocVQA	VQAv2
ScienceQA answer-only	0.800	0.805	0.645	0.703	0.630
ScienceQA rationale CE	0.780	0.760	0.450	0.559	0.300
ScienceQA explanation-aware	0.740	0.765	0.460	0.491	0.410
A-OKVQA answer-only	0.690	0.825	0.635	0.677	0.700
A-OKVQA rationale CE	0.665	0.790	0.645	0.764	0.600
A-OKVQA explanation-aware	0.680	0.795	0.615	0.763	0.640
ChartQA answer-only	0.420	0.415	0.660	0.831	0.680
DocVQA answer-only	0.450	0.435	0.685	0.910	0.660
VQAv2 answer-only	0.440	0.575	0.660	0.716	0.730

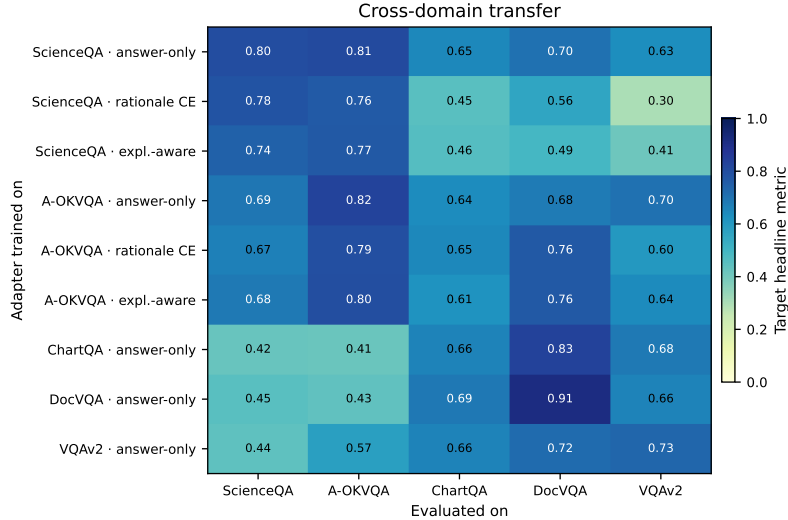


Figure 11: Cross-domain transfer heatmap. The strongest non-diagonal transfer is not always from the visually closest domain. Answer-only ScienceQA transfers strongly to A-OKVQA, while answer-only document and chart adapters transfer strongly to several open-ended targets.

5. Discussion

The results do support the main idea in some ways, but not as a simple success story. Changing the objective does change the behaviour of the model, but the claim that explanation supervision is always better than answer-only supervision is clearly not supported by the results of the completed runs. Answer-only training is the strongest on both ScienceQA and A-OKVQA. On ScienceQA, the best fixed explanation-aware run is achieved when the answer weight is low ($\alpha = 0.10$), and the length-aware control is close to the rationale-generative control, while the answer-span-only control performs worse than both of those. The same story can be seen on A-OKVQA, where explanation-aware training improves the accuracy above the rationale-generative control, but the gap is small enough that it should not be treated as statistically significant given the capped evaluation protocol.

The fixed $\alpha = 0.50$ ScienceQA run is also below the rationale-generative and answer-only controls. So the main takeaway is not just “add explanations”, but rather find the right balance of answer and rationale supervision for the domain and model.

The BLIP-2 contrastive result is also modest. On ScienceQA, the

540 contrastive-enhanced control improves the performance just slightly above
 541 the generative control (0.480 versus 0.475), but that difference is statistically
 542 insignificant given the approximate confidence intervals. The retrieval diag-
 543 nostic tells the same story, staying close to the frozen and random references.
 544 In this setup, the auxiliary image-answer InfoNCE term does not produce
 545 any stronger representation by itself. That is still a useful result, because
 546 it keeps the contrastive branch from being viewed as helpful just because it
 547 adds more seemingly useful signal to the training.

548 The domain results show why the study needs more than one benchmark.
 549 ScienceQA and A-OKVQA are used for the rationale-supervised training, be-
 550 cause they provide the gold explanations that make that training possible.
 551 ChartQA, DocVQA, and VQAv2, on the other hand, do not have gold expla-
 552 nations, and that is why those runs are used only for answer-only adaptation
 553 and domain transfer testing across structured documents, charts and natural
 554 images, rather than for rationale-supervised training. They are used to show
 555 how the same fine-tuning pipeline behaves when moved to domain where
 556 the model must answer directly without gold rationale supervision. For the
 557 stronger document-faithfulness claim we would need document-specific ra-
 558 tionale labels or a more direct OCR/layout intervention test, because the
 559 current setup does not strongly prove that the model is using the document
 560 evidence in a faithful way, even though it achieves a strong ANLS score.

561 The masking results should also be interpreted carefully. If masking some
 562 region causes the answer score to drop, that means that the model is sensitive
 563 to that part of the image. However, that does not automatically mean that
 564 the generated explanation is faithful. DocVQA and ChartQA show higher
 565 masking drift because their answers often depend on local text, labels, or
 566 layout, even though those runs use answer-only training. On ScienceQA and
 567 A-OKVQA, the masking drift is low and does not change much with the
 568 explanation-aware training, even though the rationale metrics do improve.
 569 That shows that the explanation quality and the visual sensitivity are not
 570 perfectly correlated. That is why, masking drift should be considered as
 571 an evidence-sensitivity test rather than as a proof of faithful explanation
 572 generation.

573 The cleanest positive result is seen in the scale comparison. Holding the
 574 ScienceQA explanation-aware objective fixed and moving from Qwen2-VL-
 575 2B to Qwen2-VL-7B improves accuracy from 0.765 to 0.885, ROUGE-L from
 576 0.643 to 0.738, and BLEU from 0.493 to 0.637. However, this does not mean
 577 that the 2B runs are not useful, because the smaller model makes the broader

sweep of configurations affordable. It does suggest that some of the weaker 2B results are capacity-related rather than only objective-related.

Transfer is mixed too. Answer-only supervision on ScienceQA transfers well to A-OKVQA, ChartQA, DocVQA, and VQAv2, while rationale-generative and explanation-aware training on ScienceQA perform worse on those targets. On A-OKVQA however, the rationale-generative and explanation-aware runs transfer better to DocVQA than the answer-only run, while the answer-only run transfers better to VQAv2. The DocVQA adapter is strong in-domain and also transfers well to ChartQA even though the visual format is quite different, which suggests that the model learns some useful structured, text-heavy visual reading skills that can be carried from documents to charts. Overall, these results show that the transfer depends on the trained objective and the dataset domain of both source and target.

5.1. Limitations

The main limitation is compute. The single-GPU setting keeps the study realistic for a small project, but it limits the number of seeds, model sizes, and full-dataset runs. The second limitation is rationale availability. ScienceQA and A-OKVQA provide gold explanations. DocVQA, ChartQA, and VQAv2 do not provide the same kind of rationale supervision, so any teacher-generated rationale experiment on those datasets would need to be reported separately from gold-rationale training. The third limitation is metric quality. BLEU and ROUGE-L can miss valid explanations that use different wording. Masking drift is closer to visual evidence use, but it depends on the masked regions and can be noisy for dense documents or charts. The fourth limitation is the capped protocol. The completed runs use capped train/evaluation subsets and one seed per full configuration. This keeps the work feasible and auditable, but it means small differences should not be over-interpreted. Finally, the execution logs validate the produced outputs, but they are not enough for precise wall-clock or peak-memory benchmarking. These limits define how far the claims in this paper should go.

5.2. Future directions

The first follow-up is to repeat the strongest configurations with more seeds and larger evaluation splits. A second step is to test newer open multimodal backbones while keeping the same objective-isolation protocol. Another useful direction is rationale annotation for document and chart

614 datasets, because those are the domains where faithful explanations would
615 be especially valuable. The masking test should also become more causal.
616 Instead of only hiding image regions, future work could edit document text,
617 chart marks, or image regions in controlled ways and measure whether the
618 answer changes as expected. It would also be useful to compare supervised
619 explanation-aware training with preference-based or reinforcement-learning
620 methods that reward grounded reasoning directly. For practical deployment,
621 future work should track latency, memory, and calibration together with ac-
622 curacy and faithfulness.

623 6. Conclusion

624 This paper studied how the training objective affects vision-language
625 reasoning under a fixed, resource-constrained fine-tuning budget. The ex-
626 periments compared zero-shot references, BLIP-2 generative and contrastive-
627 enhanced alignment, and Qwen2-VL answer-only, rationale-generative, and
628 explanation-aware training across document, chart, science, commonsense,
629 and natural-image benchmarks. The evaluation reports answer quality, ratio-
630 nale quality, and visual sensitivity together. The strongest supported pattern
631 is careful Qwen2-VL task adaptation: use rationale supervision where gold
632 rationales exist, and use answer-only fallback where they do not. BLIP-2
633 contrastive-enhanced training was tested directly, but its gain over BLIP-2
634 generative training is small and the retrieval diagnostic is weak. The se-
635 lected runs improve on all five datasets, but the interpretation is different
636 by domain and supervision type. The faithfulness analysis also shows an
637 important split: better rationale text does not automatically mean stronger
638 visual evidence use. Explanation-aware training improves rationale metrics
639 in some settings, but it does not consistently increase masking-based visual
640 sensitivity. The scale result is clearer, with Qwen2-VL-7B improving the
641 fixed ScienceQA explanation-aware setup over Qwen2-VL-2B. The transfer
642 matrix shows that alignment behaviour remains target-domain dependent.
643 Overall, the study supports a cautious conclusion: supervised rationales can
644 help resource-constrained VLMs, but only when answer-only controls, do-
645 main shift, and evidence-sensitivity diagnostics are reported together.

646 Declarations

647 **Author responsibility.** The authors wrote, reviewed, and edited the
648 manuscript and take full responsibility for the final content, results, and

649 claims.

650 **Use of generative AI.** Generative AI tools were used only for language
651 editing, code assistance, and consistency checks. All scientific decisions, ex-
652 periment selection, result interpretation, and final manuscript approval re-
653 main the responsibility of the authors.

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655 **Conflict of interest.** The authors declare no conflict of interest.

656 **Data, code, and artifacts.** The experiments use public datasets cited
657 in the paper. The code, configuration files, generated tables, figures, exe-
658 cution logs, and claim-to-evidence artifacts are maintained with the project
659 repository and can be made available with the submitted supplementary ma-
660 terial.

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